Pax Virentia

An Economic Simulation Model for Sustainable Pax Silica Implementation Using Machine Learning Models

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CHAPTER 1

INTRODUCTION

A digital twin is a computer-based model that represents how a real system behaves, allowing users to simulate different situations before applying changes in real life (Fuller et al., 2022). This technology was first used in industries such as aviation and manufacturing, but it is now applied in other fields, including environmental management, because it allows decision-makers to anticipate outcomes before committing resources (Botín-Sanabria et al., 2022). In this study, the digital twin framework is realized through Pax Virentia—a proposed web-based simulation tool designed to serve as a virtual model of a water treatment plant.

Central to this initiative is the evaluation of Pax Silica Implementation across local water treatment infrastructure. Pax Silica represents an advanced water treatment technology under consideration by several facilities to improve operational efficiency and lower direct processing expenses. However, operating any water treatment process demands substantial electrical power, and much of the Philippines' energy grid relies on fossil-fuel-based sources (Department of Energy Philippines, 2023; International Energy Agency, 2023). Consequently, adopting Pax Silica introduces potential environmental costs alongside its technical advantages.

To evaluate these complex dynamics, Pax Virentia is structured as an Economic Simulation Model. This computational system quantifies both financial parameters (such as energy expenditure and operational costs) and physical inputs to model prospective adoption scenarios. By simulating various operational parameters, the tool enables decision-makers to project resource requirements and monetary impacts prior to real-world deployment.

Furthermore, the model addresses critical Ecological Trade-Offs, also referred to within the Ecological-Economic Impact Simulation (EEIS) framework. Increased electricity consumption correlates directly with elevated greenhouse gas emissions, which contribute to land and marine ecosystem stress (Zawartka et al., 2020). Without rigorous assessment, the economic gains of Pax Silica could be undermined by severe environmental degradation.

By integrating economic forecasting with environmental metrics, Pax Virentia seeks to achieve a truly Sustainable operational model. International bodies emphasize that development and technological adoption must not compromise environmental integrity (United Nations Environment Programme, 2021; World Bank, 2022). Pax Virentia directly supports this principle by identifying operational thresholds where Pax Silica can be deployed effectively without compromising ecological standards.

The core problem is that water treatment plants currently decide whether to adopt Pax Silica based mainly on cost and performance, without a clear way to assess its environmental impact. No existing tool allows decision-makers to view, in one place, how much energy Pax Silica will use, how much carbon it will emit, or how it may affect nearby ecosystems. As a result, agencies such as the Department of Environment and Natural Resources (DENR), Local Water Units, and the Department of Energy (DOE) lack a shared, evidence-based basis for evaluating Pax Silica.

To address this gap, Pax Virentia provides regulators, decision-makers, and the public with a comprehensive, data-driven platform. By simulating both economic outcomes and ecological trade-offs, the project equips stakeholders to make balanced, sustainable decisions regarding Pax Silica implementation.

Background of the Study

Access to safe and adequately treated water remains a foundational service for public health, economic activity, and environmental stability in the Philippines. To meet growing demand and tightening regulatory standards, water treatment facilities are continually evaluating new treatment technologies that promise improved efficiency and lower operating costs. Pax Silica has emerged as one such technology under active consideration by several facilities, offering the potential to streamline treatment processes and reduce direct processing expenses.

This potential, however, is inseparable from an underlying structural condition of the Philippine energy sector: a significant share of national electricity generation continues to rely on fossil fuel sources (Department of Energy Philippines, 2023; International Energy Agency, 2023). Because water treatment operations are energy-intensive by nature, any technology that increases the electrical load of a plant — even while improving processing efficiency — carries a corresponding, and often unexamined, environmental cost in the form of greenhouse gas emissions and downstream ecosystem stress (Zawartka et al., 2020).

At present, the decision to adopt Pax Silica within a given facility is made primarily on the basis of cost and operational performance. Regulatory and oversight bodies such as the Department of Environment and Natural Resources (DENR), individual Local Water Units, and the Department of Energy (DOE) do not have access to a shared, localized, and evidence-based tool that translates a facility's operational choices into measurable environmental outcomes. This absence mirrors a broader pattern observed internationally, where economic development decisions risk being made without adequately accounting for environmental integrity (United Nations Environment Programme, 2021; World Bank, 2022).

It is within this context that the researchers propose Pax Virentia: An Economic Simulation Model for Sustainable Pax Silica Implementation Without Ecological Trade-Offs (EEIS). Building on digital twin technology — an approach already applied across industries to simulate outcomes before committing physical resources (Fuller et al., 2022; Botín-Sanabria et al., 2022) — Pax Virentia is intended to give Philippine water treatment stakeholders a way to evaluate Pax Silica's costs and benefits in a single, evidence-based platform prior to full-scale implementation.

Conceptual Framework

The conceptual framework of Pax Virentia follows the Input-Process-Output (IPO) model, illustrating how raw data on energy, cost, and environmental factors are transformed through Pax Virentia's simulation components into outputs that support sustainable decision-making on Pax Silica implementation.

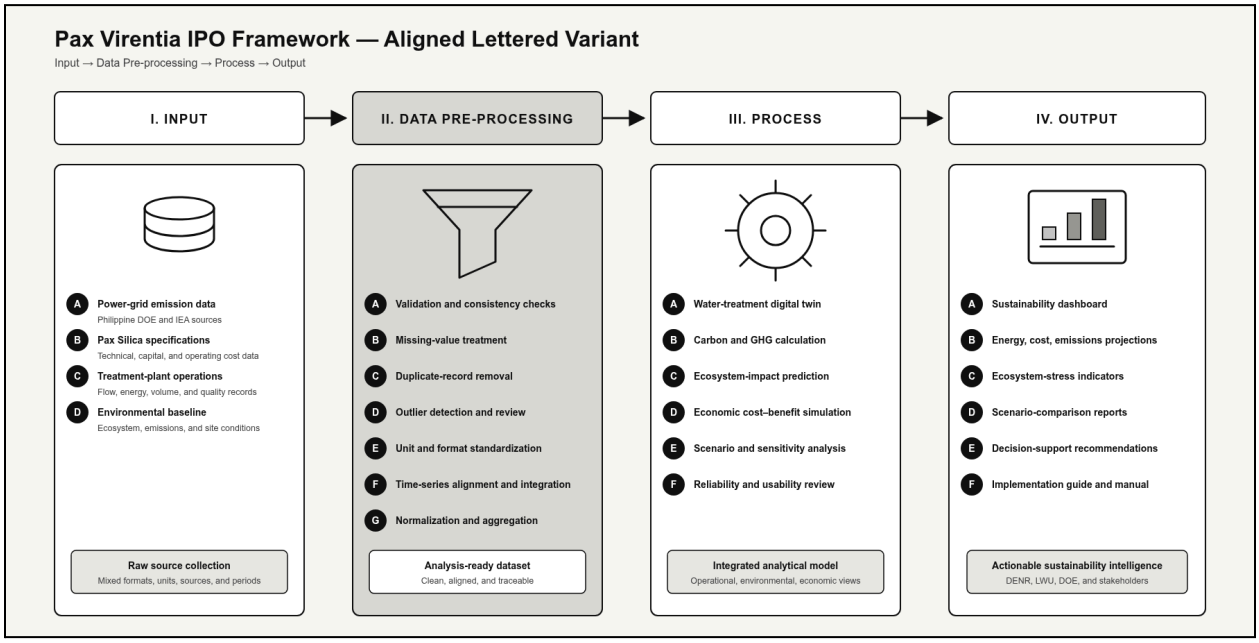


Figure 1. Conceptual Framework of Pax Virentia (Input-Process-Output Model)

As shown in Figure 1, the conceptual framework of the study is anchored on the Input-Process-Output (IPO) model, which describes how raw data is transformed into the decision-support outputs delivered by Pax Virentia.

Input. The input phase consists of the data required to model the operational, economic, and environmental effects of implementing Pax Silica. This includes Philippine power-grid emission data from the Department of Energy and the International Energy Agency. Localized emission factors are necessary because generic values may not accurately represent the country's energy mix. The phase also includes the technical specifications and cost data of Pax Silica, operational data from the water treatment plant under study—such as energy consumption, treatment volume, processing requirements, and water-quality measurements—and baseline ecosystem data for assessing potential downstream stress on terrestrial and marine environments.

Data Pre-processing. Before the collected data are used in the computational models, they undergo data pre-processing to improve their quality, consistency, and suitability for analysis. This phase includes validating records, checking for missing or inconsistent values, removing duplicate entries, identifying and reviewing outliers, and correcting formatting errors. Measurements from different sources are converted into standardized units and formats, while time-series records are aligned according to common reporting periods. Where appropriate, data are normalized, aggregated, and integrated into a unified dataset. These procedures produce a clean, traceable, and analysis-ready dataset for the digital twin, emissions calculator, ecosystem-impact predictor, and economic simulation modules.

Process. The process phase covers the core computational components of Pax Virentia. A digital twin model represents the behavior of the water treatment plant, allowing different

operating and Pax Silica adoption scenarios to be simulated before real-world implementation. The model supplies relevant operational data to a carbon or greenhouse-gas emissions calculator and an ecosystem-impact predictor. Together, these components translate energy consumption and treatment activities into measurable environmental effects. An economic simulation module also calculates the projected implementation costs, operating expenses, and potential savings associated with each scenario. Because the platform is intended for regulators and plant operators, this phase includes reliability, accuracy, and usability evaluation to determine whether its calculations and outputs are sufficiently consistent, understandable, and useful for decision-making.

Output. The output phase represents the practical results made available to stakeholders. These include the Pax Virentia sustainability dashboard, which presents energy, cost, emissions, and environmental information in an accessible format; projected energy consumption, implementation costs, savings, and carbon emissions for a selected adoption scenario; ecosystem-stress indicators that identify potential environmental effects; and scenario-comparison reports that support evaluation of alternative operating conditions. The system will also provide an implementation guide and user manual to help Department of Environment and Natural Resources staff, Local Water Unit officers, Department of Energy regulators, and plant operators use the platform correctly. Together, these outputs provide regulators, plant operators, and the public with a shared, evidence-based basis for determining whether and how Pax Silica should be adopted.

Statement of Objectives

General Objective

To develop and evaluate **Pax Virentia**, a web-based Economic Simulation Model that quantifies operational cost savings alongside power-grid carbon emissions and ecosystem stress indicators for Pax Silica deployment in Philippine water treatment plants

Specific Objectives

Specifically, this project aims to:

1. **Apply the chosen software development methodology** to systematically structure the planning, web development, and deployment phases of the Pax Virentia platform

2. **Integrate** the system's core computational modules into a central Sustainability Dashboard:

   a. **Digital Twin Model**: Simulates water treatment plant energy consumption across operational operational scenarios
   b. **Carbon Emissions Calculator**: Converts energy usage into greenhouse gas (CO_2e) emissions using Philippine power-grid emission factors from the Department of Energy (DOE) and International Energy Agency (IEA)
   c. **Ecosystem-Impact Predictor**: Estimates land and marine ecosystem stress indicators resulting from elevated emissions
   d. **Economic Simulation Module**: Calculates financial trade-offs between operational cost savings and increased electrical expenditure

3. **Test and evaluate** the platform using sample plant datasets to verify predictive accuracy, system reliability, and usability.

4. **Produce a standardized implementation guide and user manual** tailored for staff at the Department of Environment and Natural Resources (DENR), Local Water Units (LWU), and Department of Energy (DOE).

Significance of the Study

This study is expected to benefit the following groups:

a. **Regulatory Agencies (DENR, DOE).** Pax Virentia provides these agencies with a localized, evidence-based simulation tool to evaluate Pax Silica proposals against consistent, Philippine-specific environmental and energy data, supporting more defensible and consistent regulatory decisions.
b. **Local Water Units and Water Treatment Plant Operators.** Facility operators gain a decision-support tool that models the economic and environmental consequences of adopting Pax Silica before committing capital or operational resources, reducing the risk of costly, poorly informed decisions.
c. **The Public.** By making the trade-offs of Pax Silica implementation visible through an accessible dashboard, the study supports greater transparency and public understanding of decisions that affect shared environmental resources.
d. **The Academic Community and Future Researchers.** This study contributes a documented, localized application of digital twin and decision-support system methodology to the Philippine water treatment sector, an area not yet well represented in the literature, and provides a foundation that future researchers may extend, validate, or adapt to other treatment technologies and facilities.
e. **The Researchers.** Undertaking this project builds the group's practical competence in simulation modeling, environmental data analysis, and software development

methodology, directly supporting their preparation for further technical and research work.

Scope and Delimitation

This study covers the design, development, and evaluation of Pax Virentia as a web-based Economic Simulation Model applied to a single representative Philippine water treatment plant scenario. The Digital Twin Model simulates the plant's energy consumption across defined operational scenarios, including variations in flow rate, pump operation hours, mixing intensity, and chemical dosage, based on Pax Silica's technical specifications and standard operational parameters. This simulated energy profile feeds the Carbon Emissions Calculator, which converts consumption into carbon dioxide-equivalent (CO2e) emissions using Philippine power-grid emission factors from the DOE and IEA rather than generic or foreign benchmarks. These emissions figures inform the Ecosystem-Impact Predictor, which projects indicative land and marine ecosystem stress levels through predictive modeling rather than direct environmental measurement. In parallel, the Economic Simulation Module calculates the financial trade-offs between operational cost savings and increased electrical expenditure. All four modules converge in the Sustainability Dashboard, designed primarily for DENR staff, Local Water Unit officers, and DOE regulators, with a simplified view for public access. The study also covers testing the platform's reliability, predictive accuracy, and usability against sample plant datasets using a recognized software quality framework such as ISO/IEC 25010, and producing an implementation guide and user manual for its intended users.

The study is delimited in the following respects. First, Pax Virentia does not integrate physical sensors, SCADA systems, or IoT hardware; all data used are drawn from publicly

available datasets and modeled scenarios rather than live field measurements. Second, its outputs are decision-support indicators only and do not substitute for a formal Environmental Impact Assessment or carry DENR-certified regulatory standing. Third, its predictive accuracy is bounded by the quality and recency of publicly available Philippine grid emission factors, does not claim plant-specific calibration beyond the modeled scenario, and refining this accuracy is identified as future work rather than a current deliverable. Fourth, the study evaluates Pax Silica exclusively and is not designed or validated for other treatment technologies or facility types. Fifth, testing uses sample datasets and a single representative scenario rather than multiple real-world plants, so findings on reliability and usability are indicative rather than generalizable across all facilities. Sixth, the system's interface and documentation are developed in English only, without localized-dialect support. Finally, the system is developed within a single academic semester by a four-member student group using cloud, AI-assisted, and hosting resources within the project's estimated budget, which bounds the depth of testing and stakeholder consultation achievable during the study period.

Definition of Terms

Carbon (GHG) Emissions. Greenhouse gases, primarily carbon dioxide (CO2) and its equivalents, released as a byproduct of electricity generation and consumption; used in this study as a quantifiable metric linking a water treatment plant's energy use to its environmental impact.

Decision Support System. A computer-based system that provides evidence-based information, models, or projections to assist decision-makers in evaluating options and outcomes before committing to a course of action.

Digital Twin. A computer-based model that represents how a real system behaves, allowing users to simulate different situations and anticipate outcomes before applying changes in real life (Fuller et al., 2022).

Ecological Trade-Offs. The environmental costs — such as increased carbon emissions or ecosystem stress — that may accompany the operational or economic benefits of a given technology or process.

Ecosystem Stress Indicator. A projected measure, generated by Pax Virentia, that estimates the potential effect of a water treatment plant's emissions on nearby land and marine ecosystems.

Ecological-Economic Impact Simulation (EEIS). The framework underlying Pax Virentia that jointly models the economic and ecological consequences of Pax Silica implementation.

Pax Silica. An advanced water treatment technology under consideration by several Philippine water treatment facilities, valued for its potential to improve operational efficiency and reduce direct processing costs, but whose broader environmental trade-offs are not yet systematically assessed.

Pax Virentia. The web-based Economic Simulation Model proposed in this study, designed to model the energy consumption, carbon emissions, and ecosystem impact associated with Pax Silica implementation in Philippine water treatment plants.

Sustainability Dashboard. The visual, user-facing component of Pax Virentia that presents energy, emissions, and ecosystem-impact outputs in an accessible format for regulators, decision-makers, and the public.

Water Treatment Plant. A facility responsible for processing raw water into water suitable for its intended use, through operations including pumping, mixing, sedimentation, and chemical treatment.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Digital technologies and environmental assessment models have increasingly been used to support decisions involving complex systems where economic performance must be considered alongside environmental consequences. In particular, digital twins, predictive models, life-cycle assessment, and environmental decision-support systems provide ways to represent physical systems, evaluate alternative scenarios, and communicate their consequences to decision-makers. These approaches are relevant to water and wastewater treatment because such facilities require substantial energy and resources, while their operation may also generate environmental impacts. The literature reviewed in this section is organized thematically around the technological and analytical foundations of Pax Virentia: An Economic Simulation Model for Sustainable Pax Silica Implementation Without Ecological Trade-Offs (EEIS). The discussion focuses on five interconnected themes: digital twins and system modeling; digitalization of water-treatment systems; energy consumption and carbon-emission assessment; environmental and ecosystem decision support; and economic simulation, sustainability assessment, and system implementation. Together, these themes establish the technical basis for developing a tool that can integrate operational, environmental, and economic information into a single decision-support platform.

Digital Twins and Sustainability-Oriented System Modeling

Digital twins provide an important foundation for simulation-based decision support because they allow physical systems to be represented and analyzed in a virtual environment. Sahu and Upadhyay (2024) described digital twins as virtual representations of real-world

objects or systems that can reflect their behavior and status throughout their life cycle. Their review noted that the technology has expanded beyond manufacturing into environmental management and sustainability, particularly because digital twins can support simulation and synchronization between physical and digital environments. Similarly, Sharma et al. (2022) reviewed the development of digital twins across different applications and emphasized their potential for real-time monitoring, simulation, optimization, and forecasting. However, the authors also identified persistent challenges, including the absence of universal digital-twin standards, differences between application domains, security concerns, and the limited availability of performance measures. These observations are important for Pax Virentia because the proposed system is not simply intended to display water-treatment information but to represent plant behavior through a simulation model. Consequently, the literature suggests that the effectiveness of Pax Virentia will depend not only on its ability to simulate a treatment plant but also on clearly defining its system boundaries, data requirements, assumptions, and evaluation measures.

The sustainability dimension of digital twins is further supported by Trienens et al. (2024), who conducted a systematic literature review examining how digital twins can contribute to sustainability throughout a system's life cycle. Their review identified opportunities for using digital twins to optimize products, improve resource use, and develop new business models while also examining existing challenges and applications. Rather than treating digital twins solely as tools for improving operational efficiency, the study positions them as technologies that can connect engineering decisions with broader sustainability objectives. Ali et al. (2025) similarly examined digital twins in relation to circular economy and sustainable-development goals, emphasizing the importance of reliable information for resource circulation, reuse,

recycling, and reduction of waste and carbon footprints. Their work also highlighted the need for supporting technologies and cooperation among organizations, researchers, and policymakers. These studies collectively indicate that a sustainability-oriented digital twin should account for more than a single performance indicator. For Pax Virentia, this supports the integration of operational variables such as energy use with environmental and economic indicators so that alternative treatment scenarios can be examined according to their broader consequences.

The relationship between digital twins and sustainability also requires attention to the limitations of simulation-based approaches. Tzachor et al. (2022) examined the potential of digital twins for advancing the Sustainable Development Goals while emphasizing the difficulties of applying them to socio-technical and socio-ecological systems. Their perspective identified the need for digital twins to be designed as inclusive, reliable, and responsible simulations rather than assuming that a virtual model automatically produces valid sustainability decisions. This is particularly relevant to Pax Virentia because its proposed ecosystem-impact component involves environmental consequences that cannot necessarily be represented by a single physical measurement. The study therefore reinforces the importance of acknowledging assumptions and uncertainties when translating complex environmental relationships into a computational model. In relation to the research questions of the present study, this literature provides a basis for examining not only whether the digital twin functions technically but also whether its outputs are sufficiently reliable and understandable for intended users.

Digital Twins and Water-Treatment Systems

The application of digital twins specifically to water and wastewater treatment provides a more direct technical foundation for Pax Virentia. Rodríguez-Alonso et al. (2024) developed a

digital-twin platform for wastewater treatment plants using a microservices architecture designed for edge-computing implementation. Their framework combined process modeling and simulation, machine-learning techniques, and information from building information models to support operation, maintenance, and decision-making. The study demonstrates that digital twins can be structured as integrated software platforms rather than isolated simulation models. This is relevant to Pax Virentia because the proposed system similarly requires multiple components—including the digital twin, environmental calculations, predictive functions, and visualization—to operate together. The study therefore supports the feasibility of using a modular software architecture for representing water-treatment processes and connecting simulation outputs to decision-support functions.

Wang et al. (2024) further examined digital twins specifically in wastewater treatment through a technical review. Their work considered the role of digital twins in representing treatment processes and connecting operational data with digital models. The study reflects the increasing use of digital technologies to improve the monitoring, modeling, and management of wastewater-treatment systems. This is particularly relevant to Pax Virentia because a water-treatment digital twin requires a sufficiently detailed representation of the system to allow users to explore changes in operational conditions. At the same time, the review reinforces the importance of data availability and model development when attempting to move from a conceptual digital twin toward a usable decision-support system. Thus, the literature supports the proposed digital-twin component while also reinforcing the research question concerning what additional information may be required to improve Pax Virentia's predictions across different treatment plants.

Kurniawan et al. (2024) approached water-treatment digitalization from a broader sustainability perspective. Their review examined the use of Internet of Things technologies, artificial intelligence, and machine learning to improve wastewater-treatment operations, with particular attention to resource utilization, energy consumption, water quality, predictive maintenance, and carbon-neutrality goals. The authors also discussed evaluation indicators such as mean absolute percentage error, root mean square error, and coefficient of determination for assessing digital solutions, while identifying cost, complexity, and data-security concerns as implementation challenges. These findings closely correspond to Pax Virentia's need to combine process simulation with prediction and environmental assessment. More importantly, the study provides support for the present research question concerning system evaluation because it demonstrates that digital water-treatment technologies should be assessed using measurable performance indicators rather than functionality alone.

Energy Consumption, Carbon Emissions, and Environmental Trade-Offs

Energy consumption is particularly important when assessing the environmental consequences of water-treatment technologies because electricity demand can contribute substantially to operational greenhouse-gas emissions. Zib et al. (2021) examined the operational carbon footprint associated with energy consumption in the U.S. drinking-water and wastewater sectors. Their analysis of 76 wastewater utilities and 64 drinking-water utilities found that grid electricity was the largest contributor to operational greenhouse-gas emissions per unit of water and estimated average carbon footprints of 0.38 kg CO_2e/m^3 for wastewater and 0.46 kg CO_2e/m^3 for drinking water. The study also emphasized the value of examining emissions over shorter time periods rather than relying exclusively on aggregate annual measurements. Although these values are specific to the United States and should not be directly transferred to Philippine

treatment plants, the study establishes an important relationship between water-treatment energy demand and carbon emissions. For Pax Virentia, this supports the need to calculate emissions from simulated electricity consumption while using locally appropriate Philippine electricity-generation or grid-emission data.

The predictive modeling of treatment-plant energy consumption provides another foundation for Pax Virentia's energy component. Harrou et al. (2023) developed a data-driven approach for predicting energy consumption in wastewater-treatment plants using data augmentation, feature selection, and deep-learning models. Their analysis of data from a Melbourne wastewater treatment plant found that incorporating augmented and lagged energy-consumption data improved predictive performance, with LSTM and BiGRU models achieving mean absolute percentage errors of 1.36% and 1.436%, respectively. The study demonstrates that energy demand in treatment facilities can be modeled using historical and operational data rather than treated as a fixed value. This is relevant to Pax Virentia because its energy-consumption estimator may need to account for changing operational conditions rather than relying on a single constant consumption rate. At the same time, the study highlights the importance of obtaining sufficiently detailed historical data, directly supporting the research question concerning additional information needed to improve predictions across different treatment plants.

Arshad et al. (2025) extended the use of digital twins to carbon-emission management by developing a digital-twin framework for life-cycle assessment in manufacturing. Their framework combined simulation, real-time data exchange, and carbon-emission monitoring to represent changing production conditions. A case study involving aluminum casting demonstrated the ability of the framework to track energy consumption and carbon emissions

through simulation and monitoring, although the authors noted that AI-based prediction and optimization remained future work. The study is relevant to Pax Virentia because it demonstrates how energy data can be connected to carbon-emission calculations within a digital-twin environment rather than being treated as separate analytical processes. This supports the proposed integration between the digital twin and carbon-emissions calculator in Pax Virentia. However, because the study focuses on manufacturing rather than water treatment, its framework should be considered a methodological reference rather than direct evidence of how a water-treatment plant should be modeled.

The Philippine energy context is particularly important when environmental calculations are intended for local decision-making. Hameed et al. (2025) examined two long-term transition pathways for the Luzon power grid using a mixed-integer nonlinear programming model and a carbon-trading mechanism. Their results showed differences between pathways involving renewable energy, carbon capture and storage, negative-emission technologies, and carbon trading, with the modeled pathway incorporating carbon capture producing lower electricity prices and stronger alignment with renewable-energy targets. The study demonstrates the value of scenario-based optimization for examining interactions between energy systems, emissions, and economic outcomes within the Philippine context. Its results should not be interpreted as direct inputs for Pax Virentia's treatment-plant model, but its use of Philippine energy assumptions and scenario comparison provides methodological support for developing locally grounded environmental and economic simulations.

A similar connection between environmental outcomes and economic modeling can be found in Shen and Zhao's (2022) study of carbon-reduction policies in China. Using a computable general equilibrium model, the authors compared carbon trading, carbon taxation,

and a combined policy approach. Their findings indicated that the combined policy scenario produced more favorable macroeconomic outcomes than implementing a single carbon-reduction mechanism, while the authors recommended relatively moderate carbon-tax rates to avoid excessive economic pressure. Although the Chinese policy environment differs substantially from that of the Philippines and the results therefore cannot be directly applied to Pax Virentia, the study demonstrates how computational simulation can be used to compare alternative environmental-policy scenarios before implementation. This supports the economic-simulation aspect of Pax Virentia, particularly its objective of allowing users to compare environmental consequences alongside economic outcomes.

Integrated Environmental Sustainability and Ecosystem Impact

Assessing sustainability requires consideration of multiple environmental dimensions because improving one indicator may worsen another. Julian et al. (2026) demonstrated this issue in their evaluation of circular brine-treatment systems through an integrated Water Footprint Assessment and Life Cycle Assessment framework. The researchers compared different treatment configurations and found that indirect water use associated with energy consumption strongly influenced the blue-water footprint, while pollutant loads affected the grey-water footprint. Importantly, their water–carbon trade-off analysis showed that reducing carbon emissions does not necessarily result in lower water consumption. This finding is highly relevant to Pax Virentia because the proposed system is specifically intended to identify situations in which an operational or economic improvement may produce an environmental trade-off. The study therefore provides a strong conceptual basis for evaluating Pax Silica through multiple indicators instead of declaring a scenario sustainable based only on cost, energy consumption, or carbon emissions.

Rashid et al. (2023) similarly examined the application of life-cycle assessment to wastewater treatment and emphasized the importance of evaluating environmental impacts across the treatment system rather than focusing on a single operational stage. Their review illustrates how LCA can be applied to identify environmental burdens associated with wastewater-treatment processes and compare alternative treatment approaches. This perspective complements the digital-twin approach because a simulation can represent operational changes while life-cycle assessment provides a framework for interpreting their broader environmental consequences. For Pax Virentia, this supports the inclusion of environmental indicators alongside economic and operational variables. However, because the proposed system is a simulation and decision-support tool rather than a full LCA study, the literature should be used to guide the selection and interpretation of environmental indicators rather than imply that Pax Virentia itself constitutes a complete LCA.

The ecosystem dimension of environmental decision-making is further supported by Reynolds et al. (2023), who described the latest features of the Ecosystem Management Decision Support System (EMDS). The system incorporates geographic information systems, logic processing, Bayesian inference, decision modeling, workflows, and cloud computing to support ecosystem-management decisions. Its architecture demonstrates how multiple forms of environmental information can be combined to produce structured decision support rather than leaving environmental assessment as a purely descriptive exercise. This is relevant to Pax Virentia's proposed ecosystem-impact predictor because the system similarly seeks to translate environmental information into decision-oriented outputs. The study also demonstrates that environmental decision-support systems can combine different analytical methods, which may inform the architecture of Pax Virentia as its ecosystem component develops.

Environmental Visualization and Decision Support

The usefulness of environmental simulation depends not only on the accuracy of its calculations but also on how effectively its results are communicated. Riffat et al. (2023) reviewed biodiversity and ecosystem-service dashboards used in landscape and urban planning and found that dashboards can communicate complex ecosystem information to decision-makers. Their analysis of 12 dashboard applications showed substantial variation in indicators, functions, visual design, and outputs, covering a wide range of biodiversity and ecosystem-service measures. The findings indicate that dashboards can act as an interface between complex environmental information and practical planning decisions. This is directly relevant to Pax Virentia because the proposed platform is intended to consolidate energy, carbon, ecosystem, and economic information into a single dashboard. The study therefore supports the use of data visualization not merely as a presentation feature but as a mechanism for making complex sustainability information more accessible to intended users.

Lee et al. (2022) provided empirical support for visualization-based environmental decision support by evaluating a system designed for environmental impact assessment. The authors noted that conventional environmental impact assessments are often presented through text and tables that may be difficult for the public to understand, and their proposed system used visualization to support stakeholder decision-making. Their evaluation found high usability, particularly among users with stronger computational skills and environmental expertise, and suggested that the system could improve communication between experts and the public while supporting environmental literacy. These findings are important to Pax Virentia because the intended users include regulators and other decision-makers who may have different levels of technical expertise. Consequently, the study supports the inclusion of understandable

visualizations and usability evaluation in Pax Virentia, directly relating to the research question concerning whether the system is reliable, accurate, and easy to use.

Economic Simulation, Sustainability Assessment, and Decision-Making

The literature indicates that environmental decisions become more meaningful when technical performance is connected to economic consequences. The study by Hameed et al. (2025) is particularly relevant because it demonstrates how energy-transition scenarios can be evaluated according to both environmental and economic criteria in the Philippine setting. Their model compared alternative pathways while considering electricity prices, carbon emissions, renewable-energy targets, technological feasibility, and carbon-trading revenues. This type of scenario comparison is conceptually consistent with Pax Virentia's objective of evaluating whether operational benefits justify associated environmental costs. Likewise, Shen and Zhao (2022) demonstrated that environmental policies can produce different economic consequences depending on the policy configuration used in the simulation. Together, these studies suggest that decision-support models should allow users to examine alternative scenarios rather than produce a single predetermined recommendation. For Pax Virentia, this provides a foundation for presenting alternative treatment configurations and comparing their economic and environmental outcomes.

The integration of multiple sustainability indicators is also demonstrated by Julian et al. (2026), whose combined water-footprint and life-cycle assessment framework revealed trade-offs that could have been overlooked if only one environmental indicator had been considered. This is particularly significant for Pax Virentia because its central research problem involves the possibility that an economically beneficial water-treatment process may generate

additional environmental burdens. The study's findings support the development of a multi-criteria assessment approach in which energy consumption, emissions, resource use, and other environmental effects are evaluated together. In this sense, Pax Virentia can build upon existing sustainability-assessment approaches while adapting them to a web-based simulation and decision-support environment. Such integration also provides a stronger basis for recommendations because the system can identify trade-offs instead of treating economic efficiency as equivalent to sustainability.

System Development, Evaluation, and Implementation

The reviewed literature also emphasizes that developing an environmental decision-support system requires more than implementing computational functions. Sharma et al. (2022) identified the lack of standardized digital-twin frameworks and performance metrics as continuing challenges, while Kurniawan et al. (2024) highlighted cost, complexity, and data-security concerns in digitalized wastewater treatment. Rodríguez-Alonso et al. (2024) demonstrated one approach to structuring a water-treatment digital twin through a modular software architecture, while Lee et al. (2022) showed the importance of evaluating usability in environmental decision-support systems. Taken together, these studies indicate that Pax Virentia should be evaluated across several dimensions, including technical correctness, predictive accuracy, usability, and the consistency of its outputs. This directly supports the third research question concerning reliability, accuracy, and ease of use. It also provides a basis for the development of an implementation guide and user manual because a system intended for government and institutional use must communicate its assumptions, procedures, limitations, and interpretation of results clearly.

The need for appropriate documentation and contextualization is particularly important when applying models outside their original setting. The studies reviewed above were conducted across different countries and domains, including manufacturing, wastewater treatment, environmental planning, China, and the Philippines. Their findings demonstrate the usefulness of digital twins, predictive models, and environmental simulation, but they also show that model inputs and assumptions are highly dependent on the system being studied. For example, Zib et al. (2021) produced carbon-footprint estimates based on the U.S. energy system, whereas Hameed et al. (2025) specifically modeled the Philippine Luzon power grid. Therefore, Pax Virentia should avoid treating international coefficients or performance values as universally applicable. Instead, the literature supports the need for Philippine-specific datasets, documented assumptions, validation procedures, and clearly defined limitations, which directly addresses the fifth research question concerning additional information required to improve predictions across different water-treatment plants.

Pax Silica and Institutional Context

The policy context surrounding the term “Pax Silica” also requires careful clarification. The U.S. Department of State (2026) describes Pax Silica as an international initiative focused on AI and supply-chain security, covering areas such as critical minerals, energy inputs, advanced manufacturing, semiconductors, AI, and technology infrastructure. The second Pax Silica Summit resulted in additional signatories and cooperation related to AI opportunities, supply-chain security, advanced manufacturing education, and related initiatives. This source is relevant to the broader technological and economic context of the term but does **not** establish Pax Silica as a water-treatment method. Therefore, if the capstone uses “Pax Silica” to refer to a separate water-treatment technology, the project must clearly define that technology and provide

an appropriate technical source for it rather than relying on the U.S. Department of State document.

Government institutional literature can nevertheless help establish the intended environment in which an environmental decision-support tool may be used. The Department of Environment and Natural Resources (DENR, n.d.) document provided by the researchers describes the agency's institutional functions and responsibilities concerning environmental and natural-resource management. Such institutional information is useful for contextualizing the proposed role of government users in an environmental decision-support system. However, because the supplied DENR document is an institutional reference rather than an empirical study of digital twins, water treatment, or environmental simulation, it should not be presented as evidence that validates the technical design of Pax Virentia. Instead, it can support the discussion of the intended institutional context, particularly when explaining why documentation, implementation procedures, and user guidance are necessary for government-oriented software.

RRL Synthesis

The reviewed literature establishes several interconnected foundations for the development of Pax Virentia. First, research on digital twins demonstrates their potential to represent physical systems, support simulation and monitoring, and connect operational information with sustainability objectives. Studies by Sahu and Upadhyay (2024), Sharma et al. (2022), Trienens et al. (2024), Ali et al. (2025), and Tzachor et al. (2022) collectively show that digital twins can support sustainability-oriented decision-making while also requiring clear system boundaries, reliable data, validation, and performance measures. Research specifically addressing water and wastewater treatment further demonstrates that digital twins can integrate

process models, machine learning, operational data, and decision-support functions. These studies therefore establish the technological foundation for the digital-twin component of Pax Virentia.

Second, the literature demonstrates the importance of treating water-treatment sustainability as a multi-dimensional problem rather than evaluating operational performance alone. Zib et al. (2021) established the relationship between water-sector energy consumption and operational greenhouse-gas emissions, while Harrou et al. (2023) demonstrated that treatment-plant energy demand can be predicted using data-driven models. Julian et al. (2026) further showed that environmental indicators can move in different directions, demonstrating the existence of water–energy–carbon trade-offs. Research on environmental dashboards and decision-support systems also indicates that complex environmental information can be made more useful through visualization and stakeholder-oriented interfaces. Together, these studies support the integration of energy, carbon, ecosystem, economic, and visualization components within a single decision-support environment.

Despite these advances, the literature does not provide a single system that directly addresses the specific research problem of Pax Virentia. Existing studies generally focus on individual components, such as digital twins for wastewater treatment, energy prediction, carbon-emission monitoring, life-cycle assessment, environmental dashboards, or economic policy simulation. The literature also demonstrates that models are highly dependent on their datasets, system boundaries, geographic context, and assumptions. This leaves a need for a system that integrates these components into a unified simulation and decision-support framework while adapting its calculations to Philippine conditions. Pax Virentia seeks to address this gap by combining a water-treatment digital twin with energy-consumption estimation,

carbon-emission calculation, ecosystem-impact prediction, economic assessment, visualization, and recommendations. The proposed evaluation, documentation, and identification of additional data requirements further respond to the implementation and reliability concerns identified across the reviewed literature.

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